



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations
 December 2024
CC1104-2 – DATA COMMUNICATIONS
 (For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Which protocol is used to transfer files over the Internet?
 A) SMTP B) FTP
 C) HTTP D) SNMP
- Which technology is used for wireless local area networks (WLAN)?
 A) Ethernet B) Bluetooth
 C) Wi-Fi D) Token Ring
- What does a DHCP server do?
 A) Manages network security B) Assigns IP addresses to devices
 C) Routes data packets D) Provides DNS services
- Which of the following is a characteristic of analog signals?
 A) Discrete values B) Continuous waveforms
 C) High fidelity D) Digital representation
- What is the function of the network layer in the OSI model?
 A) Segmenting data B) Path determination and logical addressing
 C) Encryption of data D) Physical addressing
- Which layer in the OSI model handles data formatting and encryption?
 A) Presentation layer B) Session layer
 C) Application layer D) Transport layer
- Which of the following is a common encryption method?
 A) SHA-1 B) TCP
 C) IP D) SMTP
- What does QoS stand for in data communication?
 A) Quality of Service B) Quantity of Signals
 C) Quick Operating System D) Query of Servers
- Which of the following protocols is used for dynamic routing?
 A) RIP B) SMTP
 C) FTP D) HTTP
- What type of addressing is used in a LAN?
 A) IP addressing B) MAC addressing
 C) URL addressing D) Hostname addressing
- Which of the following protocols is used for retrieving emails?
 A) SMTP B) FTP
 C) POP3 D) HTTP
- Which topology is most resilient to a single point of failure?
 A) Star B) Bus
 C) Ring D) Mesh
- What is the main advantage of using a mesh topology?
 A) Easy to install B) High redundancy
 C) Low cost D) Simple configuration
- What is a common use for NAT (Network Address Translation)?
 A) Enhancing bandwidth B) Providing a firewall
 C) Connecting multiple devices to the Internet D) Data compression

15. Which of the following refers to the time taken for a packet to travel from the source to the destination?
- A) Bandwidth
C) Throughput
B) Latency
D) Jitter
16. What does the term "broadcast" mean in networking?
- A) Sending data to a specific device
C) Sending data to the Internet
B) Sending data to all devices in a network
D) Sending data to multiple networks
17. Which protocol is used to resolve IP addresses to MAC addresses?
- A) ARP
C) DNS
B) RARP
D) DHCP
18. What is a disadvantage of using a bus topology?
- A) High cost
C) Limited cable length
B) Difficult to troubleshoot
D) Single point of failure
19. Which data link layer protocol is commonly used for Ethernet networks?
- A) HDLC
C) IEEE 802.3
B) PPP
D) ATM
20. What does the acronym VLAN stand for?
- A) Virtual Local Area Network
C) Variable Local Area Network
B) Virtual Long Area Network
D) Virtual Linked Area Network

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the various functionalities of the Physical layer.	8	L*2	1	1
	b) What is a peer-to-peer process? Explain.	8	L2	1	1
2.	a) Encode the following bits using Manchester and differential Manchester line coding schemes. 010011	6	L3	1	2
	b) Explain Pulse Code Modulation with an example.	10	L2	2	1
3.	a) We need to send data 3 bits at a time at a bit rate of 3 Mbps. The carrier frequency is 10 MHz. Calculate the number of levels (different frequencies), the baud rate, and the bandwidth.	5	L3	2	2
	b) Write a short note on Constellation Diagrams.	5	L2	2	1
	c) Show the constellation diagrams for an ASK (OOK), BPSK, and QPSK signals.	6	L3	2	1
Unit – II					
4.	a) Explain the Hamming code with its encoder and decoder structure.	8	L3	3	2
	b) Construct the code word for the data word (1001) using CRC-3 (1011). Also draw the CRC hardware for (1011).	8	L3	3	2
5.	a) What is spread spectrum? Explain FHSS.	8	L2	3	2
	b) What is Random Access protocol? What are the procedures a station must be followed in accessing the medium? Explain.	8	L2	4	2
6.	a) How does a station access the channel in pure ALOHA? Explain.	8	L2	4	2
	b) Explain CSMA with an example.	8	L2	4	1
Unit – III					
7.	a) Explain IEEE standard for LANs.	8	L2	5	1
	b) Briefly explain Gigabit Ethernet.	8	L2	5	1
8.	a) Explain MAC layers in IEEE 802.11 standard.	8	L2	5	1
	b) Explain the three phases of circuit-switched network that required for actual communication.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
